A DETERMINANT BOUND OF 498 FOR THE TENTH DISCRIMINANT

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Abstract. We address the challenge of obtaining a small explicit upper bound for a hypothetical tenth imaginary quadratic discriminant of class number one, without using Heegner's methods or the known classification. We prove the computer-assisted bound $D \leq 498$ for every negative fundamental discriminant $-D$ with class number one. The proof constructs a nonzero interpolation determinant from polynomial values and derivatives at algebraic integers of small absolute value. Inert primes force its norm to have a large divisor, while a spectral estimate bounds the same norm from above. A rank theorem allows the determinant to be formed using nearly the minimum number of points permitted by the dimensions of the matrix. This reduces the upper bound enough to exclude $D = 499$. The numerical comparisons are evaluated with directed rounding.

1. Introduction

Let $-D < 0$ be a fundamental discriminant, let $F_D = \mathbb{Q}(\sqrt{-D})$, and write $\mathcal{O}_D$ for its ring of integers. Gauss's class number one problem asks when $\mathcal{O}_D$ has unique factorization, or equivalently when its ideal class number $h(-D)$ is one. Heegner [3], Baker [1] and Stark [5] proved that this occurs exactly for

$$D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Here we prove a weaker bound by a different method.

Theorem 1.1. *Let $-D < 0$ be a fundamental discriminant with $h(-D) = 1$. Then $D \leq 498$.*

The challenge is to obtain as small an explicit upper bound as possible for a hypothetical tenth discriminant, without relying on Heegner's methods or the known classification [7]. We approach this problem through the determinant method developed in recent work on Landau–Siegel zeros [2, 6]. In particular, the bound in Theorem 1.1 is obtained from determinant inequalities; we do not exclude the remaining discriminants by a finite test of quadratic residues.

1.1. **Overview of the proof.** Suppose that $h(-D) = 1$ with $D > 498$. We will form a matrix by evaluating polynomials in (z, w) and their z-derivatives at pairs $(z, \bar{z})$ with $z \in \mathcal{O}_D$. If this matrix has full row rank, it has a nonzero square minor Δ. Since its entries lie in $\mathcal{O}_D$, the norm $N = \Delta\bar{\Delta} = |\Delta|^2$ is a positive integer. We will bound N from above using the absolute values of the entries, and show that it is divisible by an integer larger than this upper bound. This contradiction will exclude the proposed value of D. The choice of polynomials and evaluation points must therefore serve both estimates while ensuring that the matrix has full row rank.

To see how the class number assumption enters, suppose that a prime p is not inert. There is then an ideal of norm p, which must be principal because $h(-D) = 1$. Its generator has norm p

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This work extends an externally verified finding made by autonomous rStation agents. See the section on AI use for details.

and cannot be a rational integer. But every non-rational element of $\mathcal{O}_D$ has norm at least $D/4$, so every prime $p < D/4$ must be inert. At such a prime, conjugation agrees with Frobenius modulo p, giving $z^p \equiv \bar{z} \pmod{p\mathcal{O}_D}$. Thus the polynomial $z^p - w$ takes values divisible by p at every point $(z, \bar{z})$ with $z \in \mathcal{O}_D$. By expressing the rows of our matrix in terms of powers of $z^p - w$, we obtain prime powers dividing N.

This change of basis must remain within the chosen polynomial space. We use the monomials $z^r w^s$ with $r + Ks < R$, where K and R are parameters. For $p \geq K$, replacing z^p by w does not increase $r + Ks$, so the triangular space of exponents permits all the required row operations. Taking several derivatives at each point gives more columns from the same set of lattice points. Differentiation reduces the resulting divisibility, but by an amount that can be bounded explicitly in terms of the derivative order. We also use inert primes above $D/4$: the same norm argument bounds the number of non-inert primes in an interval by the number of elements of $\mathcal{O}_D$ whose norms lie there.

For the upper bound, we choose the n lattice points of $\mathcal{O}_D$ nearest to the origin. Reducing n reduces the radius of the disk containing them and hence the bounds on the polynomial values. The estimates must also account for the derivative columns at each point. These columns are far from orthogonal, so bounding each separately loses too much at the smaller values of D. Instead, we estimate their Gram matrix, retaining the saving from these correlations before combining the contributions of the different points.

It remains to determine how few points we can use without losing full row rank. If the polynomial space has dimension M and we take J derivative orders at each point, there are nJ columns, so necessarily $nJ \geq M$. We prove a sufficient condition close to this necessary one. Direct reduction modulo an auxiliary inert prime p loses information because $z^p - w$ vanishes at every reduced node. We therefore rescale both the polynomial rows and the derivative columns using their known divisibility, obtaining an integral matrix whose reduction retains this information. A hypothetical row relation then gives a polynomial in two variables with prescribed local vanishing. Applying a differential operator and taking a determinant of coefficients produces a nonzero polynomial in one variable whose zeros, counted with multiplicity, exceed its degree. This contradiction permits a smaller n and therefore improves the upper bound for N.

We apply these estimates as follows. Elementary norm arguments first reduce $D > 498$ to primes congruent to 3 modulo 8. A uniform estimate and 101 adjacent interval comparisons exclude all $D \geq 875$, leaving sixteen possible values below 875. Fifteen are excluded by fixed determinant comparisons; the last is 499. Table 1 shows the effect of the smaller node count there, with the polynomial space and derivative orders fixed. After the derivative-order contributions have been combined, the arithmetic term changes only slightly, whereas the size estimate decreases enough to make the normalized difference positive.

	Larger node count	Smaller node count
Nodes n	41 793 600	35 394 400
Proportion $M/(nJ)$	0.83333…	0.98399…
Arithmetic minus size bound	−0.12343…	+0.05119…

TABLE 1. The comparison at $D = 499$, with the polynomial space and derivative orders fixed (approximate values). The directed bounds are given in Section 6.

Section 2 introduces the determinant and the inequality we must contradict. Sections 3 and 4 develop its divisibility and size estimates. Section 5 proves the rank criterion, and Section 6 completes the numerical argument. The explicit arithmetic formulas, subsidiary spectral proofs, uniform estimates for $D \geq 875$, and numerical calculations are collected in the accompanying repository, https://github.com/dualverse-ai/tenth_discriminant.

2. The determinant comparison

2.1. **A model: an integer that is small and highly divisible.** Every nonzero integer N satisfies

$$\log|N| = \sum_p v_p(N)\log p. \tag{2.1}$$

Our integer will be $N = |\Delta|^2$ for a nonzero algebraic integer $\Delta \in \mathcal{O}_D$, which is the norm of Δ and hence a positive rational integer. The strategy is to bound $\log N$ from above by Archimedean estimates, to bound $v_p(N)$ from below at many primes, and to show that the two bounds are incompatible when $h(-D) = 1$. This subsection illustrates the comparison using elementary estimates. The rest of the paper sharpens each of its three ingredients.

The matrix. We work with polynomials in two independent variables z and w. The variable w will be specialized to the complex conjugate of the value of z, but it is treated as independent of z throughout. For $P \in \mathbb{Z}[z,w]$ define the Hasse derivatives in z, with w held fixed, by

$$P(z+t,w) = \sum_{j\ge 0} (\partial_z^{[j]}P)(z,w)\,t^j, \qquad \partial_z^{[j]}(z^r w^s) = \binom{r}{j} z^{r-j} w^s.$$

Hasse derivatives keep integer coefficients; ordinary derivatives would introduce a factor $j!$. Fix a positive integer J. Let $z_1,\dots,z_n \in \mathcal{O}_D$ be distinct *nodes*, and let $\Lambda \subset \mathbb{Z}^2_{\ge 0}$ be a finite set of exponents. The *jet matrix* has rows indexed by $(r,s) \in \Lambda$, columns indexed by pairs (i,j) with $1 \le i \le n$ and $0 \le j < J$, and entries

$$V_{(r,s),(i,j)} = (\partial_z^{[j]} z^r w^s)(z_i, \overline{z_i}) = \binom{r}{j} z_i^{\,r-j}\,\overline{z_i}^{\,s} \in \mathcal{O}_D. \tag{2.2}$$

For a set C of $|\Lambda|$ columns, write Δ_C for the corresponding maximal minor. More generally, the row of any polynomial P in the span of the monomials $z^r w^s$, $(r,s) \in \Lambda$, is the vector of its jets $(\partial_z^{[j]}P)(z_i,\overline{z_i})$. Integral unitriangular changes of the spanning polynomials leave every Δ_C unchanged.

One Frobenius calculation. The divisibility comes from the behaviour of conjugation modulo an inert prime.

Lemma 2.1. *Let p be a prime that is inert in F_D, and let $z \in \mathcal{O}_D$. Then $z^p \equiv \overline{z} \pmod{p\mathcal{O}_D}$. Consequently, if $g = z^p - w$, then*

$$g(z+t,\overline{z}) = pU(t) + t^p$$

for a polynomial $U \in \mathcal{O}_D[t]$ of degree less than p, and for all integers $m, j \ge 0$,

$$p^{\lfloor j/p\rfloor}\,[t^j]\,g(z+t,\overline{z})^m \in p^m\mathcal{O}_D. \tag{2.3}$$

Proof. The quotient $\mathcal{O}_D/p\mathcal{O}_D$ is the field of p^2 elements. Conjugation induces its nontrivial automorphism, which is the pth-power map. This gives $z^p \equiv \bar{z} \pmod{p\mathcal{O}_D}$, including when $p = 2$. Next,

$$(z+t)^p - \overline{z} = (z^p - \overline{z}) + \sum_{k=1}^{p-1}\binom{p}{k} z^{p-k}t^k + t^p,$$

and every term except t^p is divisible by p. Finally, $(pU+t^p)^m = \sum_{q=0}^m \binom{m}{q} p^{m-q}U^{m-q}t^{pq}$, and a term contributes to t^j only if $q \le \lfloor j/p\rfloor$. Each such term is divisible by $p^{m-\lfloor j/p\rfloor}$, which proves (2.3). □

Geometrically, the lemma says that, p-adically, every point $(z,\overline{z})$ lies close to the curve $z^p = w$ [6]. The polynomial g is small at every node, and its z-derivatives are small too, since $\partial_z g = pz^{p-1}$. This is why the jets differentiate only in z: a derivative in w would not preserve the divisibility.

Where the triangle comes from. To use Lemma 2.1, write each row exponent as $r = mp + a$ with $0 \le a < p$ and replace the monomial $z^r w^s$ by

$$g^m z^a w^s = z^r w^s + \sum_{i=1}^{m} (-1)^i \binom{m}{i} z^{r-pi} w^{s+i}. \tag{2.4}$$

By the Leibniz rule and (2.3), every jet of $g^m z^a w^s$ of order j, multiplied by $p^{\lfloor j/p \rfloor}$, lies in $p^m \mathcal{O}_D$. The change of basis (2.4) is unitriangular for the ordering by z-degree, but only if every exponent $(r - pi, s + i)$ on the right is again in Λ. The support must therefore be closed under the Frobenius substitution $z^p \mapsto w$.

The rectangle $\{r < R,\ s < S\}$ used by Green is not closed under this substitution: it is stopped by the ceiling $s < S$. There the substitution can be applied only $\min(\lfloor r/p \rfloor, S - 1 - s)$ times, and the p-adic credit of the row is truncated accordingly [2, Lemma 5.1]. The remedy is to grade the monomials so that the substitution never raises the degree. Fix integers $K \ge 2$ and $S \ge 1$. Give z weight 1 and w weight K, and take

$$\Lambda = \{(r, s) \in \mathbb{Z}^2_{\ge 0} :\ r + Ks < R\}, \qquad R = KS, \qquad M = |\Lambda| = \frac{R(S+1)}{2}. \tag{2.5}$$

The substitution $z^p \mapsto w$ changes the weight by $K - p$, so it never leaves Λ when $p \ge K$. Every row then receives its full credit $\lfloor r/p \rfloor$. Primes $p < K$ receive truncated credit, and that boundary is treated in Section 3.

Scaling column (i, j) by $p^{\lfloor j/p \rfloor}$ multiplies Δ_C by $p^{\sum_{(i,j)\in C} \lfloor j/p \rfloor}$. Since $p\mathcal{O}_D$ is prime and $v_p(|\Delta_C|^2) = 2 v_{p\mathcal{O}_D}(\Delta_C)$, we obtain, for every inert prime $p \ge K$,

$$v_p(|\Delta_C|^2) \ \ge\ 2 \sum_{(r,s)\in\Lambda} \Big\lfloor \frac{r}{p} \Big\rfloor \ - \ 2 \sum_{(i,j)\in C} \Big\lfloor \frac{j}{p} \Big\rfloor. \tag{2.6}$$

The row contribution, including its factor 2, is about W/p, where $W = R^2 S/3 \approx 2 \sum_\Lambda r$. The second is a column loss that is present only when derivative orders reach p. In our application J is much larger than p, so the loss is real. Section 3 shows how it combines with the factorial sizes of the Hasse coefficients.

Sizes. If every node satisfies $|z_i| \le \rho$ with $\rho \ge 1$, then each entry of row (r, s) has absolute value at most $\binom{r}{j} \rho^{r+s-j}$. Hadamard's inequality therefore gives

$$\log |\Delta_C|^2 \ \le \sum_{(r,s)\in\Lambda} 2(r+s) \log \rho \ + \sum_{(r,s)\in\Lambda} \log \sum_{(i,j)\in C} \binom{r}{j}^2 \rho^{-2j}. \tag{2.7}$$

The first term is $(1 + 1/K + O(1/S))\, W \log \rho$. Combining (2.1), (2.6) and (2.7), any nonzero Δ_C satisfies

$$\sum_{\substack{p \ge K \\ p \text{ inert}}} 2\Big(\sum_{\Lambda} \Big\lfloor \frac{r}{p} \Big\rfloor - \sum_{C} \Big\lfloor \frac{j}{p} \Big\rfloor\Big) \log p \le \ \sum_{\Lambda} 2(r+s) \log \rho + \sum_{\Lambda} \log \sum_{C} \binom{r}{j}^2 \rho^{-2j}. \tag{2.8}$$

After division by W, the left side is roughly the sum of $(\log p)/p$ over inert primes $K \le p < R$, less the column loss. Under $h(-D) = 1$ this inert sum is bounded below by the corresponding sum over all primes, minus a loss controlled by counting values of the norm form, as in Section 3. The right side is roughly $(1 + 1/K) \log \rho$ plus the derivative term. The theorem follows when the left side is larger.

This model already shows where the difficulty lies, and it points to the three refinements. First, the arithmetic credit must be computed exactly, including small primes, prime powers, and the loss from non-inert primes, bounded through the norm form (Section 3). Second, Hadamard's

inequality treats the entries of each node block as unrelated and discards their cancellation. The spectral estimates of Section 4 recover part of it. Because the nonzero minor is not identified, those estimates must hold for every column set C. Third, ρ is governed by the node count. The n nearest lattice points lie in a disk of radius about $\sqrt{n\sqrt{D}/(2\pi)}$ (Lemma 2.5), and $n \geq M/J$ is forced.

2.2. **Arithmetic of class number one.** Every element of $\mathcal{O}_D$ has the form $\alpha = (x + y\sqrt{-D})/2$ with $x, y \in \mathbb{Z}$ and $x \equiv yD \pmod 2$, and its norm is

$$N(\alpha) = |\alpha|^2 = \frac{x^2 + Dy^2}{4}. \tag{2.9}$$

The elementary reductions below follow from one observation about norms.

Lemma 2.2 (Norm obstruction)**.** *If $\alpha \in \mathcal{O}_D \setminus \mathbb{Z}$, then $N(\alpha) \geq D/4$. If $\alpha \in \mathbb{Z}$, then $N(\alpha)$ is a square.*

Proof. If $\alpha \notin \mathbb{Z}$ then $y \neq 0$ in (2.9), so $N(\alpha) \geq Dy^2/4 \geq D/4$. If $\alpha \in \mathbb{Z}$ then $N(\alpha) = \alpha^2$. □

Lemma 2.3. *Suppose $h(-D) = 1$. Then every prime $p < D/4$ is inert in F_D.*

Proof. If p is split or ramified, there is a prime ideal of norm p. It is principal, generated by some α with $N(\alpha) = p$. Since p is not a square, $\alpha \notin \mathbb{Z}$, and Lemma 2.2 gives $p \geq D/4$. □

Lemma 2.4 (Reduction)**.** *Suppose $h(-D) = 1$ and $D > 16$. Then D is a prime and $D \equiv 3 \pmod 8$.*

Proof. The ramified primes are the prime divisors of D, so by Lemma 2.3 every prime divisor of D is at least $D/4$. If D is even, then 2 ramifies and $2 < D/4$, which is impossible. Hence D is odd, and since $-D$ is a fundamental discriminant, D is squarefree with $D \equiv 3 \pmod 4$. If D were composite, its least prime factor q would satisfy $q \leq \sqrt{D} < D/4$, again impossible. So D is prime. Finally, if $D \equiv 7 \pmod 8$ then $-D \equiv 1 \pmod 8$ and 2 splits, contradicting Lemma 2.3. Therefore $D \equiv 3 \pmod 8$. □

For the proof of Theorem 1.1 we assume $D > 498$ from now on, so D is a prime congruent to 3 modulo 8, and both 2 and every odd prime below $D/4$ are inert. Since $D \equiv 3 \pmod 4$, the integer cutoff $p < D/4$ is equivalently $p \leq (D-3)/4$. We write $\omega = (1 + \sqrt{-D})/2$, so that $\mathcal{O}_D = \mathbb{Z}[\omega]$. In these coordinates $\mathrm{N}(x_0 + y\omega) = (x_0 + y/2)^2 + Dy^2/4$; it is the same norm form as (2.9), with $x = 2x_0 + y$.

2.3. **Nearest nodes.** Our nodes are the lattice points of $\mathcal{O}_D \subset \mathbb{C}$ nearest to the origin. Let $z_1, z_2, \ldots$ be the elements of $\mathcal{O}_D$ listed with $|z_1| \leq |z_2| \leq \cdots$, with ties broken arbitrarily. By (2.9), $\mathcal{O}_D$ is the union of the horizontal rows

$$\left\{ \tfrac{x}{2} + \tfrac{y}{2}\sqrt{-D} : \ x \equiv yD \pmod 2 \right\}, \qquad y \in \mathbb{Z}.$$

The rows are spaced $\sqrt{D}/2$ apart vertically, and within each row the points form a translate of $\mathbb{Z}$. The covolume of $\mathcal{O}_D$ is therefore $\sqrt{D}/2$.

Lemma 2.5 (Nearest-node disk)**.** *Put $\delta = \sqrt{D+4}/4$. For every $i \geq 1$,*

$$|z_i| \ \leq \ \rho_i := \sqrt{\frac{i\sqrt{D}}{2\pi}} + \delta. \tag{2.10}$$

Proof. Every point of the plane is within vertical distance $\sqrt{D}/4$ of a lattice row, and within horizontal distance $1/2$ of a point on that row. Thus the lattice has covering radius at most δ. For $z \in \mathcal{O}_D$, let $V(z)$ be its Voronoi cell, the set of points of $\mathbb{C}$ at least as close to z as to any other lattice point. These cells are translates of one another, they cover $\mathbb{C}$, and they overlap only on their boundaries. Each has area $\sqrt{D}/2$. Every point of $V(z)$ lies within δ of z.

Let $\rho > \delta$. If $|\zeta| \le \rho - \delta$ and $\zeta \in V(z)$, then $|z| \le |\zeta| + \delta \le \rho$. Hence the closed disk of radius $\rho - \delta$ is covered by the cells of lattice points of modulus at most ρ, and comparing areas gives

$$\#\{z \in \mathcal{O}_D : \ |z| \le \rho\} \ \ge \ \frac{2\pi(\rho-\delta)^2}{\sqrt{D}}.$$

With $\rho = \rho_i$ the right side equals i, so at least i lattice points have modulus at most ρ_i. This gives $|z_i| \le \rho_i$. □

We use Lemma 2.5 in two ways. It supplies the radius $\rho = \rho_n$ in the Archimedean estimates. It also controls residues, through the following consequence.

Corollary 2.6 (Distinct residues)**.** *Let p be a prime with $2\rho_n < p$. Then $z_1, \dots, z_n$ have distinct residues modulo $p\mathcal{O}_D$.*

Proof. A nonzero element $p\beta$ of $p\mathcal{O}_D$ has modulus $p\sqrt{N(\beta)} \ge p$. The difference of two distinct nodes is nonzero and has modulus at most $2\rho_n < p$. □

At $D = 499$, with $n = 35\,394\,400$, Lemma 2.5 gives $2\rho_n < 22\,491.5$. This is far below every candidate auxiliary prime in $(160\,000, 170\,000]$, so the residue hypothesis of the rank theorem holds whichever of those primes is inert.

2.4. **The inequality to be proved.** We now give the comparison in a form that separates the algebra from the numerical estimates. The notation introduced here will be used throughout the paper. Write

$$W = \frac{R^2 S}{3}, \qquad W_1 = 2\sum_{(r,s)\in\Lambda} r, \qquad W_2 = 2\sum_{(r,s)\in\Lambda} s. \tag{2.11}$$

The factor W is a convenient normalization: the main logarithmic terms in both estimates have size $W \log D$. Direct summation gives

$$\frac{W_1 + W_2}{W} = 1 + \frac{1}{K} + \frac{3(K-1)}{2KS} + \frac{K-5}{2KS^2}. \tag{2.12}$$

We retain the two finite-size terms. In particular, the coefficient of $\log\rho$ is not replaced by its limit $1 + 1/K$ in the numerical comparisons.

Choose M columns C for which the square submatrix V_C is nonsingular, and put $\Delta_C = \det V_C$. The rank argument will supply such a choice; it will not specify the columns. The arithmetic estimates therefore have to hold for every possible choice C. They take the form

$$W\mathcal{A} - 2\sum_{(i,j)\in C} b(j) \ \le \ \log|\Delta_C|^2. \tag{2.13}$$

Here $\mathcal{A}$ is a column-independent lower bound assembled from local divisibilities, and $b(j)$ is the debit for using a derivative of order j. Both depend on the chosen parameters. Section 3 describes their construction and states the local estimate used below.

The main part of $b(j)$ is the first-prime-power sum $\sum_q \lfloor j/q \rfloor \log q$. Legendre's formula shows that subtracting it from $\log(j!)$ leaves higher-prime-power savings, with additional refunds from

the treatment of non-inert primes. This remaining function is nearly linear in j; we subtract a slope and bound its finite remainder. Set

$$F(j) = \log(j!) - b(j), \qquad H(j) = j\sigma - F(j), \tag{2.14}$$

where σ is a slope chosen in the arithmetic calculation. Let c be any bound such that

$$2 \sum_{(i,j)\in C} H(j) \le Mc \quad \text{for every } |C| = M. \tag{2.15}$$

There are only n columns of any one order j, so this is a finite selection problem with a capacity constraint. It need not be bounded by the much cruder $2M \max_j H(j)$.

Let ρ contain all the nodes, put $u_i = z_i/\rho$, and define

$$B_{(r,s),(i,j)}(\ell) = \frac{(r)_{\underline{j}}}{R^j} e^{\ell j} u_i^{r-j} \bar{u}_i^{s}, \qquad \ell = \log(R/\rho) - \sigma. \tag{2.16}$$

Here $(r)_{\underline{j}} = r(r-1)\cdots(r-j+1)$ is the falling factorial, with $(r)_{\underline{0}} = 1$; the entry is zero when $j > r$. The exact relation to the integral matrix is

$$B_C(\ell) = \operatorname{diag}_\Lambda(\rho^{-r-s})\, V_C\, \operatorname{diag}_C(j!(\rho/R)^j e^{\ell j}). \tag{2.17}$$

Taking squared absolute determinants, substituting (2.14), and then using (2.13) gives

$$W\mathcal{A} \le (W_1 + W_2)\log\rho + \log|\det B_C(\ell)|^2 + 2\sum_C H(j).$$

This calculation explains the normalization: the factorials are cancelled exactly, rather than estimated twice.

Proposition 2.7 (Determinant comparison)**.** *Suppose that V has full row rank, that (2.13) and (2.15) hold, and that*

$$\frac{1}{W}\log|\det B_C(\ell)|^2 \le U_{\mathrm{var}} \qquad (|C| = M)$$

holds uniformly over all column selections. Then

$$\mathcal{A} \le \frac{W_1 + W_2}{W}\log\rho + U_{\mathrm{var}} + \frac{Mc}{W}. \tag{2.18}$$

In particular, a strictly larger arithmetic lower bound is impossible.

Proof. Choose a nonsingular minor and apply the preceding identities. All bounds are uniform in the choice, so no information about which minor is nonsingular is needed. □

The earlier entrywise estimates pay a factor at most $M!$ for expanding a determinant. Our numerical comparisons retain the resulting nonnegative allowance $2M\log M/W$, even when Fischer's inequality makes this allowance unnecessary. At the parameters for 499 it is less than 3×10^{-7}, much smaller than the final gap. We retain it so that the reported values can be reproduced directly. Thus define

$$\mathcal{U} = \frac{W_1 + W_2}{W}\log\rho + U_{\mathrm{var}} + \frac{Mc + 2M\log M}{W}, \qquad G = \mathcal{A} - \mathcal{U}. \tag{2.19}$$

A positive lower bound for G rules out the proposed field. There are three separate tasks: prove full rank, establish the local arithmetic estimates, and bound the same minor at the complex embedding. The auxiliary-prime scalings used to prove rank in Section 5 do not enter (2.17); after nonvanishing has been established, all size and divisibility estimates concern the original integral minor Δ_C.

It is already possible to see why the number of nodes matters. The disk estimate gives $\rho \asymp n^{1/2}D^{1/4}$ once its boundary error is small. Reducing n therefore lowers the common term in (2.19) by approximately

$$\frac{1}{2}\left(1+\frac{1}{K}\right)\log\frac{n_{\mathrm{old}}}{n_{\mathrm{new}}}.$$

At the two node counts in Table 1, this is about 0.085. The remaining terms also change: there are fewer copies of each derivative order, and the proportion $\theta = M/(nJ)$ of available columns that must be selected increases. We call this proportion the *occupancy*. Section 4 exploits these restrictions. In the actual calculation the combined gain is enough to change the sign of G.

3. Divisibility of the minor

We write q for a general prime, reserving p for the auxiliary prime in Section 5. The basic Frobenius argument in Section 2.1 gives a row contribution from each inert prime and a loss depending on the derivative orders selected. We now strengthen the row contribution, first by using the conjugate variable and then by retaining the small primes. Finally we explain how class number one controls the losses from non-inert primes.

Recall that for an inert prime $q \geq K$, the integral row basis is $P_{r,s} = z^{r \bmod q}(z^q - w)^{\lfloor r/q \rfloor} w^s$. Writing v_q for the valuation normalized by $v_q(q) = 1$, the calculation in Lemma 2.1 gives

$$v_q(\partial_z^{[j]} P_{r,s}(z_i, \bar{z}_i)) \geq \max(0, \lfloor r/q \rfloor - \lfloor j/q \rfloor). \tag{3.1}$$

Each determinant term uses every row and every selected column once. Hence

$$v_q(\Delta_C) \geq \sum_{\Lambda} \lfloor r/q \rfloor - \sum_{C} \lfloor j/q \rfloor. \tag{3.2}$$

The right side can be negative, in which case it is merely a weak bound. Its usefulness emerges after summing over primes: the column loss combines with the factorials in the Hasse derivatives, as in (2.14).

3.1. Additional divisibility in the conjugate variable.

Differentiation in z leaves polynomials in w untouched. Such polynomials can contribute additional divisibility at no derivative cost. We use the following elementary tower. Define

$$T_1(w) = w^{q^2} - w, \qquad \nu_a = \frac{q^{2a}-1}{q^2-1}, \qquad T_{a+1}(w) = T_a(w)^{q^2} - q^{\nu_a(q^2-1)} T_a(w).$$

These are monic polynomials of degrees q^{2a}. Inductively,

$$T_a(\bar{z}_i) \in q^{\nu_a}\mathcal{O}_D. \tag{3.3}$$

Indeed, if $T_a(\bar{z}_i) = q^{\nu_a} b$, then the next value equals $q^{q^2\nu_a}(b^{q^2} - b)$, which is divisible by $q^{q^2\nu_a+1} = q^{\nu_{a+1}}$.

Write $s = \sum_{a\geq 0} d_a q^{2a}$ in base q^2, and put

$$A_s(w) = w^{d_0} \prod_{a\geq 1} T_a(w)^{d_a}.$$

It is monic of degree s, and (3.3) gives

$$v_q(A_s(\bar{z}_i)) \geq \sum_{a\geq 1} d_a \nu_a = \sum_{a\geq 1} \left\lfloor s/q^{2a} \right\rfloor.$$

Replacing w^s by $A_s(w)$ in the previous basis preserves triangularity. We obtain the stronger row estimate

$$v_q(\partial_z^{[j]}(z^{r \bmod q}(z^q - w)^{\lfloor r/q \rfloor} A_s(w))(z_i, \bar{z}_i)) \geq \max(0, \lfloor r/q \rfloor - \lfloor j/q \rfloor) + \sum_{a\geq 1} \left\lfloor s/q^{2a} \right\rfloor. \tag{3.4}$$

This is a local statement for any nodes in the order; it has no rank hypothesis.

The row contributions are finite sums over the triangle. Their exact floor-sum formulas are given in the supplementary notes, Section S2.

3.2. **Small primes and non-inert primes.** For an inert prime $q < K$, replacing z^q by w increases weighted degree. The number of replacements in a row is therefore bounded by

$$h_q(r,s) = \min\left(\lfloor r/q \rfloor, \lfloor (R-1-r-Ks)/(K-q) \rfloor\right). \tag{3.5}$$

There is still a substantial contribution: summing this function over the triangle yields

$$2\sum_{\Lambda} h_q(r,s) = W_2 = \frac{R(S^2-1)}{3}.$$

One way to verify the identity is to sum the sizes of its superlevel sets. The condition $h_q \geq a$ is equivalent, after writing $r = qa + r'$, to $r' + Ks < R - Ka$, and its number of solutions is $K(S-a)(S-a+1)/2$. Summing for $1 \leq a < S$ gives the displayed value.

The small-prime estimate used in the computation improves this elementary bound by combining $z^q - w$, $w^q - z$, and $z^{q^2} - z$. The first factor can lose a power of q only after spending at least q orders of differentiation; the latter two can lose one after spending one order. Optimizing where each type is used is a finite triangular row calculation. The resulting finite lower bound is recorded in the supplementary notes, Section S5, together with its cut parameter. In particular, $q = 2$ is included. Working in the maximal order is essential here, since 2 is inert when $D \equiv 3 \pmod 8$.

The arithmetic calculation also uses primes above $(D-3)/4$, where inertness is no longer guaranteed. Class number one still supplies useful information. If q splits or ramifies, an ideal of norm q is principal, so

$$q = \mathrm{N}(x_0 + y\omega) = \left(x_0 + \frac{y}{2}\right)^2 + \frac{D}{4}y^2, \qquad y \neq 0. \tag{3.6}$$

At a fixed norm cutoff there are only finitely many such pairs. One can therefore start with an all-prime estimate and subtract an upper bound for the contribution of possible non-inert primes, using these norm representations. This deduction uses principality; it does not test the quadratic character at a particular discriminant.

Counting all norm values as possible primes overestimates the loss. Unique factorization allows some of those values to be removed in advance. Products of two or three nonrational factors have composite norm, and the corresponding primitive products can be counted with their multiplicities controlled. The product counts used for this subtraction are recorded in the supplementary arithmetic notes. At the finite comparisons in Section 6, only the two- and three-factor corrections are used. The upper bound for possible losses is decreased by a rigorously bounded contribution of these composite values. Thus this correction *increases* $\mathcal{A}$; it is not an additional subtraction of prime divisibility.

For orientation, the resulting arithmetic term has the form

$$\mathcal{A} = A_{\mathrm{all}} - L_{\mathrm{norm}} + C_2 + C_3. \tag{3.7}$$

The first term sums inert-prime row credits as a provisional all-prime bound; L_{norm} pays for norm values that might be non-inert primes; C_2 and C_3 restore credits paid unnecessarily for certified composite values. Split and ramified primes also have their own local conjugate-variable contributions. These, and the derivative-order refunds, are included in the arithmetic lower bound. This matters numerically near $D = 499$.

3.3. **The arithmetic estimate used below.** We use the following consequence of the local estimates, in a form that continues to hold when the number of nodes is reduced. The normalization is $\log|\Delta_C|^2/W$: all prime-sum and norm-loss terms in $\mathcal{A}$ are divided by W, while $b(j)$ is the column debit in logarithmic units before this division.

Proposition 3.1 (Local arithmetic estimate)**.** *Let $-D$ be a fundamental discriminant with $h(-D) = 1$ at one of the finite comparisons below* 875 *in Section 6, with the parameters specified there. Let $z_1, \dots, z_n$ be the n nearest elements of $\mathcal{O}_D$, and let V be the matrix (2.2) with $0 \le j < J$ and $\theta = M/(nJ) \le 1$. Define $\mathcal{A}, \sigma, c, b(j)$ by the local arithmetic formulas in the supplementary notes, Section S5, including the norm-product corrections and derivative-order refunds. Then (2.13) and (2.15) hold for every set C of M columns with $\Delta_C \neq 0$.*

Here Δ_C is a minor of the original integral Hasse matrix V. The bound on D will follow by comparing these local estimates with the size estimate of Section 4. The preceding arguments explain the constructions behind the estimates. The supplementary notes give the finite formulas and their derivations. Every quantity depending on n is evaluated at the node count in use. This includes the number of columns of each order, and hence the least possible sum of their first-power floors and the largest possible remainder sum.

The higher-prime-power saving can be seen directly from Legendre's formula:

$$\log(j!) = \sum_q \sum_{a \ge 1} \lfloor j/q^a \rfloor \log q.$$

The debit $\sum_q \lfloor j/q \rfloor \log q$ from (3.2) cancels its $a = 1$ part exactly. The remaining terms, together with the split-prime refunds, form $F(j)$ in (2.14). Sorting $H(0), \dots, H(J-1)$ in decreasing order, say as $H_{(1)}, \dots, H_{(J)}$, and writing $M = nt + r$ with $0 \le r < n$, yields the exact capacity bound

$$\sum_C H(j) \le n \sum_{a=1}^{t} H_{(a)} + r H_{(t+1)}. \tag{3.8}$$

When $r = 0$ the final term is omitted. The right side gives an admissible $Mc/2$. Grouping the floor functions into intervals on which their quotients are constant gives a finite evaluation of the right side. The resulting bound for c includes the remainder at every derivative order.

4. An upper bound for every column selection

We turn to the complex absolute value of the minor. An entrywise bound followed by Hadamard's inequality is enough for a large discriminant bound. At the smaller values of D considered here, however, it loses too much information about the derivative columns at a single node. Those columns are far from orthogonal. Our estimate keeps them together, bounds their Gram matrix, and then combines the node contributions.

There is a complication worth making explicit. Full row rank gives a nonsingular minor but does not tell us how many of its columns come from each node, or which derivative orders are selected. Neither the first nor the last m orders need maximize an m-column minor. The estimates below therefore allow an arbitrary subset of orders at each node, and only use the restrictions that there are at most J columns per node and M columns in total.

4.1. **Weights and node Gram matrices.** Let $B(\ell)$ be the normalized matrix in (2.16). Choose real numbers $\lambda_{r,s}$ and form

$$Y = \operatorname{diag}_\Lambda(e^{-\lambda_{r,s}/2}) B(\ell).$$

Denote by Y_i the block of J columns at node i, and write $G_i = Y_i^* Y_i$. If C_i is the set of orders selected at that node, Fischer's inequality gives

$$\log|\det B_C(\ell)|^2 \le \sum_{\Lambda} \lambda_{r,s} + \sum_{i=1}^{n} \log\det G_i[C_i, C_i]. \tag{4.1}$$

Indeed, $Y_C^* Y_C$ is a positive semidefinite Gram matrix, and its determinant is at most the product of the determinants of its diagonal blocks. The weights are free parameters, but their total cost $\sum \lambda_{r,s}$ must be restored. Some weights have negative logarithmic cost; this signed sum will be estimated separately.

Lemma 4.1 (Radial monotonicity)**.** *For fixed row weights and a fixed set of derivative orders, each principal minor of a node Gram matrix is a polynomial with nonnegative coefficients in $q = |u_i|^2$.*

Proof. Write $u_i = \sqrt{q}\, e^{\mathrm{i}\phi}$. The phase of an entry of Y_i is $e^{\mathrm{i}(r-s-j)\phi}$, so it is removed by diagonal unitary transformations on the rows and columns. By Cauchy–Binet, the principal Gram minor is a sum of squared minors of Y_i. For a row set E and order set T of equal size, such a squared minor has the form

$$e^{-\sum_E \lambda_{r,s}} \prod_{j\in T} \frac{e^{2\ell j}}{R^{2j}} \big(\det[(r)_{\underline{j}}]_{E,T}\big)^2 q^{\sum_E (r+s) - \sum_T j}.$$

If the determinant is nonzero, at least one term in its expansion is nonzero. That term matches each j to a row with $r \ge j$, so the exponent of q is nonnegative. All coefficients are nonnegative, proving the claim; the value at $q = 0$ follows by continuity. □

Thus a node can be moved outward to a radius bound before estimating its selected Gram minor. This assertion concerns principal minors, not monotonicity of individual eigenvalues. It is exactly the form needed to replace the ordered nodes by a finite set of radial bounds.

4.2. **Why negative-binomial weights help.** Fix parameters $a > 0$, $0 < v < 1$ and $\beta < 0$, and let $b = aR$. We use

$$e^{-\lambda_{r,s}} = \frac{(b)_r}{r!} v^r e^{-\beta s}, \tag{4.2}$$

where $(b)_r = b(b+1)\cdots(b+r-1)$. These are negative-binomial weights in the r variable. They are useful because their moments against falling factorials can be computed explicitly.

The triangle allows a better estimate than completing the rows to a rectangle. Write $x_+ = \max(0, x)$. At a node of squared radius $q > 0$, put

$$\delta_q = (\log q - \beta)_+, \qquad v_q = v e^{-\delta_q/K}, \qquad w = v_q q, \qquad \mathcal{B} = S e^{S\delta_q}.$$

For a fixed $r < R$, the permitted s satisfy $0 \le s < S - r/K$. Hence

$$\sum_{0\le s<S-r/K} (q e^{-\beta})^s \le S e^{(S-r/K)\delta_q}. \tag{4.3}$$

Multiplication by $e^{-r\delta_q/K}$ replaces v by v_q. We may then extend the r sum to all nonnegative integers: the added row outer products are positive semidefinite. After removing the node angle by diagonal unitary conjugation, G_i is bounded above in the positive semidefinite order by the $J \times J$ matrix

$$\mathcal{G}_{jk} = \mathcal{B}\left(\frac{e^{\ell}}{R\sqrt{q}}\right)^{j+k} \sum_{r\ge 0} \frac{(b)_r}{r!} w^r (r)_{\underline{j}} (r)_{\underline{k}}, \qquad 0 \le j, k < J. \tag{4.4}$$

The formulas at $q = 0$ are understood by continuity. In the finite radial partition used below all upper radii are positive. This completion changes the estimate, not the integral determinant being bounded. All its principal minors majorize the corresponding ones of G_i.

Here is the moment calculation behind the choice of weights. Normalize $(b)_r w^r/r!$ to have total mass one. The resulting random integer r satisfies

$$\mathbb{E}[(1+x)^r(1+y)^r] = \left(\frac{1-w}{1-w(1+x)(1+y)}\right)^b. \tag{4.5}$$

Factoring the denominator as

$$(1-w)((1-\alpha x)(1-\alpha y)-\delta_0 xy), \qquad \alpha = \frac{w}{1-w}, \quad \delta_0 = \frac{w}{(1-w)^2},$$

and expanding in x and y gives a triangular orthogonal factorization. Let P_l be the monic polynomials orthogonal for this normalized negative-binomial measure. These are the Meixner polynomials; their formulas are

$$\|P_l\|^2 = l!(b)_l \frac{w^l}{(1-w)^{2l}}, \tag{4.6}$$

$$(r)_{\underline{j}} = \sum_{l=0}^{j} \binom{j}{l} (b+l)_{j-l} \left(\frac{w}{1-w}\right)^{j-l} P_l(r). \tag{4.7}$$

The generating identity proves the constants directly. No limiting asymptotic formula for orthogonal polynomials is involved.

Choose a column scale $\tau > 0$ and set

$$\nu_j = \log(j!) + \log((b)_j) + 2j(\ell - \log(R\tau)). \tag{4.8}$$

After dividing column j by $e^{\nu_j/2}$, the factorization (4.6)–(4.7) identifies the completed matrix factor with $\sqrt{\mathcal{B}(1-w)^{-b}}$ times the matrix of composition by

$$\phi(z) = c_0 + dz, \qquad c_0 = \frac{\tau v_q \sqrt{q}}{1-w}, \qquad d = \frac{\tau\sqrt{v_q}}{1-w}, \tag{4.9}$$

on polynomials of degree less than J. The Hilbert space has orthonormal monomials $\sqrt{(b)_j/j!}\, z^j$, or equivalently reproducing kernel $(1-z\bar{w})^{-b}$. For example, the coefficient in row l, column j of composition is

$$\binom{j}{l}\sqrt{\frac{(b)_j l!}{j!(b)_l}} c_0^{j-l} d^l,$$

which is precisely the normalized factor obtained above. This converts a large derivative Gram matrix into a problem about an affine map of the disk.

4.3. **Two bounds for singular products.** We state the two estimates used in the spectral part of the computation. We prove the first here and give the second proof in Section S3 of the supplementary notes.

Lemma 4.2 (Affine composition)**.** *Suppose $c_0 \geq 0$, $d > 0$ and $c_0 + d < 1$. Set*

$$U = c_0 d, \quad V = d^2, \quad Z = 1 - c_0^2, \qquad h = \frac{2U}{Z - V + \sqrt{(1-(c_0+d)^2)(1-(c_0-d)^2)}},$$

$$\Gamma = (Z - Uh)^{-1}, \qquad \xi^2 = \frac{V + Uh}{Z - Uh}.$$

The squared singular values of composition by $c_0 + dz$ on the space with kernel $(1-z\bar{w})^{-b}$ are $\Gamma^b \xi^{2l}$, $l \geq 0$.

Proof. One has $0 \le h < 1$, $0 < \xi < 1$, and $Uh^2 - (Z - V)h + U = 0$. The kernel of the product of the composition operator with its adjoint is

$$[Z - U(z + \bar{w}) - Vz\bar{w}]^{-b}.$$

The disk automorphism $(z - h)/(1 - hz)$, accompanied by the multiplier $(1 - h^2)^{b/2}/(1 - hz)^b$, induces a unitary operator, as is checked by substituting in the reproducing kernel. Conjugating the displayed kernel by this unitary and using the quadratic equation for h gives Γ^b times the kernel of the dilation operator $f(z) \mapsto f(\xi^2 z)$. Its eigenvalues on the monomials are ξ^{2l}. Taking square roots gives the singular values of the original operator. □

Compression to polynomials of degree less than J cannot increase singular values. If m orders are selected at a node, let $N_m(\tau)$ be the sum of the largest m numbers among $\nu_0, \dots, \nu_{J-1}$. Restoring the column scales and the scalar mass gives the bound

$$F_q^{\mathrm{f}}(m) = N_m(\tau) + m \log \mathcal{B} + bm \log \frac{\Gamma}{1 - w} + m(m - 1) \log \xi \tag{4.10}$$

for its logarithmic Gram determinant. This applies to every selection of m orders. The term $m(m - 1) \log \xi$ is negative and quadratic in m. It records the geometric decay of the singular values, a saving lost when all m singular values are bounded by the largest one. This explains why selecting many columns at a node can improve the estimate. The other terms, including the capacity cost N_m, still have to be paid; the final calculation accounts for all of them together. Conveniently, one may choose $0 < \xi < 1$ first and then set

$$\Gamma = \frac{1 + \xi^2 + \sqrt{(1 - \xi^2)^2 + 4w\xi^2}}{2}, \qquad \tau = \frac{\xi(1 - w)}{\Gamma \sqrt{v_q}}.$$

These choices automatically give a contracting map in (4.9).

Near $m = J$, it can be better to estimate the complementary singular product. This allows an expanding finite affine map as well.

Lemma 4.3 (Complementary singular product)**.** *Let T_J be composition by $c_0 + dz$ on polynomials of degree less than J, where $c_0 \ge 0$ and $d > 0$. Put $k = J - m$. For $0 < r_0 < 1$, define*

$$G_0 = \frac{1 + r_0^2 + \sqrt{(1 - r_0^2)^2 + 4c_0^2 r_0^2}}{2}, \qquad s_0 = \frac{dr_0}{G_0}.$$

The logarithm of the product of the largest m squared singular values of T_J is at most

$$J(J - 1) \log d + kb \log G_0 + k(k - 1) \log r_0 - N \log s_0, \tag{4.11}$$

where $N = k(2J - k - 1)$ for $s_0 \le 1$, and $N = k(k - 1)$ for $s_0 \ge 1$.

Adding $N_m(\tau) + m \log \mathcal{B} - bm \log(1 - w)$ to (4.11) bounds the same node minor as (4.10). We can take the minimum over either estimate and over any finite collection of its parameters. Each choice gives an upper bound for the original selected minor, so minimizing does not presume that a single parameter is optimal at every node.

4.4. **A bound summed over all order sets.** The preceding estimates bound a selected Gram minor through singular values. We can also bound it by summing all principal minors of the same size: each term is nonnegative, so the sum bounds any one of them, regardless of which derivative orders occur in the nonsingular minor. The explicit moments above give the following estimate for this sum. Its Schur-function proof is given in the repository notes on spectral estimates. Let

$$M_{jk}(b, w) = \sum_{r \ge 0} \frac{(b)_r}{r!} w^r (r)_{\underline{j}} (r)_{\underline{k}}, \qquad G_{jk} = \mathcal{B} \sigma_0^{j+k} M_{jk}(b, w),$$

where $b, \mathcal{B}, \sigma_0 > 0$ and $0 < w < 1$. Put $c_* = \sigma_0\sqrt{w}/(1-w)$ and $z_* = (bc_*)^{-2}$. If $bc_* > 1$, then, for $0 \le m \le J$, with $a_* = J - m$ and $s_* = ma_* + m(m-1)/2$,

$$\sum_{\substack{T\subseteq\{0,\dots,J-1\}\\ |T|=m}} \det G[T,T] \le \mathcal{B}^m (1-w)^{-bm} c_*^{2s_*} \prod_{i=0}^{m-1} i!(b)_{2a_*+i}\,(1-z_*)^{-m^2}. \tag{4.12}$$

The empty product is one. In our application $G = \mathcal{G}$ from (4.4), with $\sigma_0 = e^{\ell}/(R\sqrt{q})$. Taking the logarithm of the right side of (4.12) therefore gives a third upper bound for every selected node minor.

The reason Schur functions appear can be described without using them in the rest of the proof. The Meixner factorization and Cauchy–Binet first bound a selected minor by a gamma moment determinant. If its order set is $j_0 < \cdots < j_{m-1}$, its deviation from consecutive orders is a partition. The gamma determinant is bounded by the consecutive-order value times the square of the corresponding Schur dimension. Summing these dimensions over all partitions uses the Cauchy identity and produces $(1-z_*)^{-m^2}$. It is the sum over all order sets, rather than an assertion that one consecutive set is largest, that makes (4.12) valid. The numerical estimate takes its minimum with the two singular-product bounds.

4.5. **Combining nodes with an occupancy constraint.** Let $F_q(m)$ be any common upper bound for the logarithmic determinant of an m-column minor at squared radius q, obtained as above. For a real parameter $a_0 > 0$, choose $\Phi(q) \ge 0$ such that

$$F_q(m) \le a_0 R m + R^2 \Phi(q) \qquad (0 \le m \le J). \tag{4.13}$$

Any function Φ satisfying this inequality will suffice. Summing over nodes and using $\sum_i |C_i| = M$, (4.1) gives

$$\frac{1}{W}\log|\det B_C(\ell)|^2 \le U_{\mathrm{row}} + \frac{3a_0(S+1)}{2S} + \frac{3}{S}\sum_{i=1}^{n}\Phi(q_i), \qquad U_{\mathrm{row}} = \frac{1}{W}\sum_{\Lambda}\lambda_{r,s}. \tag{4.14}$$

This elementary use of a common slope incorporates the total number of selected columns. Bounding each node independently at its worst occupancy would discard that information.

The disk estimate supplies

$$q_i \le Q(i/n), \qquad Q(t) = \left(\frac{\sqrt{t}+\eta}{1+\eta}\right)^2, \qquad \eta = \frac{\sqrt{D+4}/4}{\sqrt{n\sqrt{D}/(2\pi)}}.$$

Partition i/n into N_t intervals. For the kth interval, first move all its nodes outward to squared radius $Q(k/N_t)$ using Lemma 4.1, and then apply (4.13) at that radius. This needs no monotonicity assumption on the chosen function Φ. Each interval contains at most $n/N_t + 1$ nodes. Substituting $n = R(S+1)/(2\theta J)$ gives

$$U_{\mathrm{var}} = U_{\mathrm{row}} + \frac{3a_0(S+1)}{2S} + 3\left(\frac{1+1/S}{2\theta(J/R)N_t} + \frac{1}{S}\right)\sum_{k=1}^{N_t}\Phi(Q(k/N_t)), \qquad \theta = \frac{M}{nJ}. \tag{4.15}$$

The nonnegativity of Φ justifies using upper bounds for the node counts. The signed row sum U_{row} remains outside that estimate.

For the finite comparisons below we take

$$(a, v, \beta, a_0) = \left(\frac{1}{5}, \frac{7}{10}, -\frac{4}{5}, \frac{3}{50}\right), \qquad (N_t, N_z, N_m) = (128, 256, 512). \tag{4.16}$$

The latter two mesh sizes are used for column-order capacity and occupancy, respectively. Whole-cell upper bounds are taken before choosing the minimum of the three estimates; endpoints alone would not bound an entire cell. The supplementary notes, Section S6, give these finite bounds.

Substitution into (4.15), with directed rounding, supplies U_{var} in the determinant comparison. At no stage is the enormous matrix itself constructed.

5. Nonvanishing with fewer nodes

The last algebraic input is full row rank of the matrix V. A dimension count requires $nJ \geq M$, but it does not prove that the rows are independent. The criterion below comes close to that count and is sufficient for all the finite comparisons between 499 and 875. The proof uses a divided-Frobenius basis to retain information lost by direct reduction modulo a prime.

Direct reduction modulo an inert prime loses information: the polynomial $z^p - w$ vanishes at every reduced node. The remedy is to divide out the powers of p whose presence we already know. After this division, a hypothetical row relation gives a polynomial in two variables with many prescribed local zeros. A differential operator converts those conditions into too many zeros of a nonzero polynomial in one variable. The two main points are to prove that this last polynomial is nonzero and to compare its degree with its orders of vanishing.

Theorem 5.1 (Sufficient rank criterion)**.** *Let p be an inert prime, and suppose that the nodes $z_1, \dots, z_n$ have distinct residues modulo $p\mathcal{O}_D$. Let $R = KS$ and*

$$K < p < R \leq p^2, \qquad 1 \leq J \leq R.$$

If $n \geq 2S$ and

$$n\left(J - \frac{RJ}{p^2} - \frac{R}{p}\right) > \frac{(R+p)^2}{2K}, \tag{5.1}$$

then the matrix V in (2.2) has full row rank.

The leading terms of (5.1) are $nJ > R^2/(2K)$, close to the necessary inequality $nJ \geq M = R(S+1)/2$. The errors are small when p/R and R/p^2 are small. For our application the choice of p must also allow distinct node residues and guarantee that at least one candidate prime is inert. These requirements will be checked together in Section 6.

5.1. **Dividing out the Frobenius factors.** Put $g = z^p - w$ and $d = \lfloor (R-1)/p \rfloor < p$. The integral basis change (2.4) gives the rows

$$g^m z^a w^s, \qquad 0 \leq m \leq d, \quad 0 \leq a < p, \quad a + Ks < R - mp. \tag{5.2}$$

At node i write

$$g(z_i + t, \bar{z}_i) = pU_i(t) + t^p, \quad U_i(t) = \frac{z_i^p - \bar{z}_i}{p} + \sum_{v=1}^{p-1} \frac{1}{p}\binom{p}{v} z_i^{p-v} t^v \in \mathcal{O}_D[t].$$

Direct reduction would replace $(pU_i(t) + t^p)^m$ by t^{mp}, discarding U_i and making layer m vanish in every column of order less than mp. Instead, divide each row in layer m by p^m and multiply each column of order $j = qp + v$, $0 \leq v < p$, by p^q. Call the resulting matrix $\widetilde{V}$. Its entries are integral. Indeed, in the binomial expansion of $(pU_i + t^p)^m$, the term indexed by q' can contribute to order j only when $q' \leq q$; after rescaling its coefficient contains $p^{q-q'}$. Modulo p, only the term $q' = q$ survives.

Write $k = \mathcal{O}_D/p\mathcal{O}_D$, $\xi_i = z_i \bmod p$, and $u_i(z) = (U_i \bmod p)(z - \xi_i)$. Thus, when $q \leq m$, the reduced entry is

$$\binom{m}{q} [t^v](u_i(\xi_i + t)^{m-q}(\xi_i + t)^a \xi_i^{ps}). \tag{5.3}$$

When $q > m$, all terms are divisible by p, and the reduced entry is zero. The basis change and all scalings are invertible in characteristic zero. It therefore suffices to prove that $\widetilde{V} \bmod p$ has full row rank over k. These scalings are used only for nonvanishing; the determinant comparison continues to concern the original matrix V.

Suppose, to the contrary, that the reduced rows satisfy a nontrivial relation with coefficients $c_{m,a,s} \in k$. Package that relation as

$$F(z,u) = \sum_{m=0}^{d} H_m(z)u^m, \qquad H_m(z) = \sum_{a,s} c_{m,a,s} z^{a+ps}. \tag{5.4}$$

Within each layer the exponents $a + ps$ are distinct, since $0 \le a < p$. Consequently $F \neq 0$. Write $e = \deg_u F \le d < p$. The triangular support gives

$$\deg H_m < E_m := \frac{p(R - mp)}{K}, \tag{5.5}$$

because $a + ps = (p/K)(a + Ks) - (p/K - 1)a < p(R - mp)/K$. The importance of this bound is its constant drop: $E_m - E_{m+1} = p^2/K > p$.

To express the local conditions on F, put

$$L_l = \min(p, \max(0, J - lp)).$$

There are exactly L_l derivative orders in the block $j = lp + v$. In characteristic p, $(\xi_i + t)^{ps} \equiv \xi_i^{ps} \pmod{t^p}$. We may therefore replace ξ_i^{ps} by $(\xi_i + t)^{ps}$ when taking any of the coefficients $[t^v]$ with $v < L_l$. Summing (5.3) against the relation coefficients gives

$$\operatorname{ord}_{z=\xi_i}\big((\partial_u^{[l]} F)(z, u_i(z))\big) \ge L_l \qquad (i = 1, \dots, n;\ l \ge 0). \tag{5.6}$$

Here $\partial_u^{[l]}$ denotes the Hasse derivative in u. Thus a relation among the scaled rows has become a family of vanishing conditions along polynomial curves $u = u_i(z)$.

5.2. **One differential operator for all nodes.** Differentiating $pU_i(t) = (z_i + t)^p - \bar{z}_i - t^p$ before reducing modulo p gives

$$u_i'(z) = z^{p-1} - (z - \xi_i)^{p-1}. \tag{5.7}$$

The first term is independent of the node, and the second vanishes to order $p - 1$ there. This suggests using the common operator

$$\mathcal{D} = \partial_z + z^{p-1}\partial_u$$

in place of differentiation along each separate curve. Its key property is

$$\mathcal{D}^p = -\partial_u \qquad \text{on } k(z)[u]. \tag{5.8}$$

Indeed, the pth iterate of a derivation in characteristic p is a derivation. It sends z to zero and u to $(p-1)! = -1$, which determines it on $k(z)[u]$.

Lemma 5.2. *The polynomials $F, \mathcal{D}F, \dots, \mathcal{D}^e F$ are linearly independent over $k(z)$. In particular*

$$W_F(z) = \det\big([u^m]\mathcal{D}^j F\big)_{0 \le j,m \le e} \tag{5.9}$$

is nonzero, and

$$\deg W_F \le \sum_{m=0}^{e} E_m - \frac{e(e+1)}{2}. \tag{5.10}$$

Proof. If these polynomials were dependent, let $r \le e$ be the first index of a dependence. Their span up to index $r - 1$ would be stable under $\mathcal{D}$: applying $\mathcal{D}$ to a rational coefficient differentiates that coefficient, and the image of the last spanning vector is in the same span by assumption. By (5.8) this space is also stable under ∂_u. It would then contain $F, \partial_u F, \dots, \partial_u^e F$. Since $e < p$, these have distinct u-degrees $e, e-1, \dots, 0$ and nonzero leading coefficients. They are independent, contradicting the dimension $r \le e$.

For the degree bound, induction using

$$[u^m]\mathcal{D}G = ([u^m]G)' + (m+1)z^{p-1}[u^{m+1}]G$$

shows that $\deg([u^m]\mathcal{D}^j F) \le E_m - j$. The first term lowers degree by one; the second has degree at most $E_{m+1} - j + p - 1 < E_m - j - 1$. In every determinant term each column index m and row index j occurs once. Adding these degree bounds proves (5.10). □

The use of $e < p$ here is essential: it prevents the successive u-derivatives from vanishing prematurely in characteristic p. The identity $\mathcal{D}^p = -\partial_u$ is what connects this elementary fact to differentiation in z.

5.3. **Counting zeros of the coefficient determinant.** Fix a node i. Translation from the basis $1, u, \dots, u^e$ to $1, u - u_i(z), \dots, (u - u_i(z))^e$ is unitriangular. It leaves W_F unchanged and replaces its entries by

$$B^{(i)}_{j,l}(z) = (\partial_u^{[l]} \mathcal{D}^j F)(z, u_i(z)).$$

We claim that

$$\operatorname{ord}_{\xi_i} B^{(i)}_{j,l} \ge \max(0, L_l - j). \tag{5.11}$$

To see this, let $\mathcal{D}_i = \partial_z + u_i'(z)\partial_u$, which differentiates along the curve. Equation (5.7) gives $\mathcal{D} - \mathcal{D}_i = (z - \xi_i)^{p-1}\partial_u$. An induction, in which each differentiation lowers the order of the error by at most one, gives

$$\mathcal{D}^j G \equiv \mathcal{D}_i^j G \pmod{(z - \xi_i)^{p-j} k[z, u]} \qquad (0 \le j < p).$$

The Hasse derivatives in u commute with both operators: their coefficients depend only on z, and ordinary and Hasse u-derivatives commute. After specializing $u = u_i(z)$, the corresponding entry is therefore congruent to the jth ordinary derivative of $(\partial_u^{[l]} F)(z, u_i(z))$, modulo $(z - \xi_i)^{p-j}$. Its order is at least $\max(0, L_l - j)$ by (5.6). Since $L_l \le p$, the error has at least that order as well. This proves the claim.

We can now finish the proof by factoring powers of $z - \xi_i$ out of rows or columns. Put $Q = \lfloor J/p \rfloor$. If $e < Q$, then $L_l = p$ for all $0 \le l \le e$. Every entry in row j of $B^{(i)}$ has order at least $p - j$, so

$$\operatorname{ord}_{\xi_i} W_F \ge \sum_{j=0}^{e} (p - j) = (e + 1)(p - e/2) > (e + 1)p/2.$$

The residues are distinct and $n \ge 2S = 2R/K$. Summing this bound over the nodes therefore gives more than $(e + 1)pR/K$ zeros counted with multiplicity, whereas (5.10) bounds the degree by at most that number. This contradicts $W_F \ne 0$.

If $e \ge Q$, every entry in column l has order at least $\max(0, L_l - e)$. Factoring column by column and using $\sum_{l=0}^{Q} L_l = J$ gives

$$\operatorname{ord}_{\xi_i} W_F \ge \sum_{l=0}^{Q} (L_l - e) = J - (Q + 1)e \ge J - \frac{RJ}{p^2} - \frac{R}{p}. \tag{5.12}$$

For the degree, write $R = dp + s_0$, where $1 \le s_0 \le p$. Then

$$2p \sum_{m=0}^{d} (R - mp) = (R - s_0 + p)(R + s_0) = R^2 + pR + s_0(p - s_0) < (R + p)^2.$$

All E_m are positive. Hence

$$\deg W_F \le \sum_{m=0}^{e} E_m \le \sum_{m=0}^{d} E_m < \frac{(R + p)^2}{2K}.$$

Summing (5.12) over the nodes and applying (5.1) again makes the total order of vanishing exceed the degree. Both cases are impossible. The reduced matrix has full row rank, and so does V, proving Theorem 5.1.

6. The numerical comparisons and the bound 498

We now assemble the proof. The finite calculations evaluate the inequalities already described; they do not construct a matrix with M rows. There are two parts to the numerical argument. A uniform estimate, supplemented by adjacent interval comparisons, excludes every $D \geq 875$. Below 875 the elementary reduction leaves sixteen possible values, each of which is excluded by a separate determinant comparison. We describe the coverage first, then give the parameters and the decisive calculation at 499.

6.1. **The range $D \geq 875$.** For this range we use the arithmetic and radial estimates in the supplementary notes, Section S7, together with a resultant rank criterion. The latter counts intersections with the horizontal and vertical rows of the lattice. If M_x bounds the number of nodes on any one horizontal lattice row and M_y bounds the number of such rows meeting the disk, its sufficient conditions are

$$nJ > R(S + M_y), \qquad \max(J, M_x) \leq R + KM_y.$$

They are verified with the disk bound, using $M_x \leq 2\rho{+}1$ and $M_y \leq 4\rho/\sqrt{D}{+}1$. The auxiliary-prime rank theorem of Section 5 is needed only for the smaller finite comparisons below.

The infinite range is controlled by an inequality with a positive logarithmic coefficient. Set $D_0 = 5\,760\,000$, $X = (D-3)/4$, and $X_0 = (D_0 - 3)/4$. With

$$K = 31, \qquad S = \left\lfloor \frac{23X}{10K} \right\rfloor, \quad R = KS, \quad J = \left\lfloor \frac{29R}{500} \right\rfloor,$$

and the node count chosen in the supplementary notes, Section S7, the arithmetic and size estimates imply

$$G_{\mathrm{tail}}(D) > 0.00031743 + 0.127829 \log(X/X_0) \qquad (D \geq D_0). \tag{6.1}$$

Here G_{tail} is the lower estimate for the normalized arithmetic-minus-size difference using the radial bound in the supplementary notes. The uniform estimates there give (6.1). The nonzero determinant comparison requires $G_{\mathrm{tail}}(D) \leq 0$, so this positive lower bound is a contradiction. It is an estimate on a whole ray: positivity at the left endpoint is accompanied by a proved positive coefficient for subsequent logarithmic growth.

The interval between 875 and D_0 is covered as follows. The gap bounds in Table 2 are rounded downward from the smallest lower endpoint in each family. Each comparison uses a closed interval, and successive intervals share an endpoint.

Range	Number of intervals	Lower gap bound
$[875, 3\,968\,539]$	28	0.00397338
$[3\,968\,539, 4\,894\,699]$	37	0.00032629
$[4\,894\,699, 5\,760\,000]$	36	0.00040940

Table 2. The finite part of the original determinant bound 875. The first family uses norm-form estimates, the second exact finite prime sums, and the third explicit prime-sum bounds of Rosser and Schoenfeld.

The complete interval schedule is included in the repository. Thus (6.1) and Table 2 exclude all $D \geq 875$ under the class-number-one hypothesis. No finite test of quadratic residues is used in this step.

6.2. **Parameters and rank for the remaining comparisons.** By Lemma 2.4, the values still possible above 498 are 499 and the fifteen primes in Table 3. For these comparisons we use $K = 43$ and a prime cutoff P. The other parameters are

$$S = \left\lfloor \frac{6P}{5K} \right\rfloor, \qquad R = KS, \qquad J = \lfloor R/8 \rfloor, \qquad M = R(S+1)/2.$$

For $D = 499$ we take $P = 312\,000\,000$ and $n = 35\,394\,400$. The remaining node counts are listed in the repository notes, Table S1.

At 499, the scales relevant to the rank argument are

$$2\rho_n < 22\,492 < 160\,000 < p, \qquad p \le 170\,000 < R = 374\,399\,968 < 160\,000^2 < p^2.$$

Thus an auxiliary prime can be large enough to separate the nodes while remaining small compared with the polynomial degree.

We first prove nonvanishing. There are 814 primes in $(A, B] = (160\,000, 170\,000]$. A non-inert prime here is split and therefore has the form $p = X^2 + Dy^2/4$, with $y \ge 1$ and $X \ge 0$ in a translate of $\mathbb{Z}$. For $D \in [D_-, D_+]$ and each y, the interval of possible X has length at most $\sqrt{B - D_- y^2/4} - \sqrt{\max(0, A - D_+ y^2/4)}$. Adding one to each length and summing over $1 \le y \le \lfloor \sqrt{4B/D_-} \rfloor$ bounds the number of non-inert primes. This gives fewer than 765, both on $[499, 503]$ and at each value in Table 3. Hence at least one of these primes is inert. The count is a finite sum of square roots, enclosed by rational bounds.

All node diameters are less than 28 928, using Lemma 2.5; this deliberately loose common bound is much smaller than A. The residues are therefore distinct for every candidate auxiliary prime. The remaining rank conditions hold uniformly for $A < p \le B$: one checks $K < A < R$, $R < A^2$, $J \le R$, $n \ge 2S$, and

$$n\left(J - \frac{RJ}{A^2} - \frac{R}{A}\right) - \frac{(R+B)^2}{2K} > 0. \tag{6.2}$$

At the smaller node count the difference in (6.2) exceeds 7.22×10^{11}. The reason for narrowing the prime interval can now be seen quantitatively. Writing $\theta = M/(nJ)$, inequality (6.2) is equivalent to

$$\theta < \frac{(1 + 1/S)(1 - R/A^2 - R/(AJ))}{(1 + B/R)^2}.$$

At the parameters for 499, the right side is approximately 0.984431 when $A = 160\,000$, just above the actual occupancy 0.983995. With $A = 150\,000$ it is approximately 0.982414, which is too small. The narrower interval improves the sufficient rank estimate while still containing more primes than the norm form can account for as non-inert.

At the other three parameter choices the respective differences exceed 4.12×10^{12}, 9.02×10^{13} and 2.95×10^{14}. Theorem 5.1 therefore proves full row rank in every case.

6.3. **The fifteen comparisons above 499.** We now evaluate Proposition 2.7, with the local arithmetic estimates of Proposition 3.1 and the spectral bound (4.15). The spectral parameters and mesh sizes are fixed as in (4.16). Table 3 gives the result; each displayed lower bound is rounded downward. The full enclosures are included in the repository. Each entry is positive and contradicts (2.18). Thus every value in the table is excluded.

6.4. **The decisive comparison at 499.** At $D = 499$ we retain $P = 312\,000\,000$, $K = 43$ and the same R, S, J as in the preceding comparison, but reduce the node count to $n = 35\,394\,400$. The arithmetic and size calculations are performed on the whole interval $[499, 503]$, using the same parameters throughout. Only 499 needs to be excluded: $503 \equiv 7 \pmod 8$ is already impossible by Lemma 2.4.

D	$P/10^6$	Lower gap bound	D	$P/10^6$	Lower gap bound
523	312	0.01913120	683	36	0.24689369
547	312	0.06770128	691	36	0.25866104
563	169	0.09816857	739	36	0.33084083
571	169	0.10589325	787	36	0.39774315
587	36	0.08275416	811	36	0.42960657
619	36	0.13882635	827	36	0.45047644
643	36	0.17968979	859	36	0.48970182
659	36	0.20911703			

TABLE 3. Positive lower bounds for $G = \mathcal{A} - \mathcal{U}$ at all fifteen primes $D \equiv 3 \pmod 8$ in $(499, 875)$.

Table 4 separates the three contributions to the comparison. The common term is

$$U_{\mathrm{common}} = \frac{W_1 + W_2}{W} \log \rho + \frac{Mc + 2M \log M}{W},$$

so that $G = \mathcal{A} - U_{\mathrm{common}} - U_{\mathrm{var}}$. The displayed arithmetic numbers are lower bounds and the size numbers are upper bounds, rounded in the safe direction. The second column is a control: it repeats the calculation with the earlier node count and all other parameters fixed.

	Earlier node count	Smaller node count
n	41 793 600	35 394 400
$\theta = M/(nJ)$, approximately	0.83333113	0.98399487
Arithmetic lower bound $\mathcal{A}$	9.99153787	9.99493844
Upper bound for U_{common}	9.71143220	9.62002864
Upper bound for U_{var}	0.40353803	0.32371122

TABLE 4. The comparison on $[499, 503]$ with $M = 1\,629\,945\,955\,088\,368$. Only the occupancy row uses approximate decimals; the other decimal entries are one-sided bounds.

The arithmetic decomposition (3.7) at the smaller node count is, approximately,

$$\mathcal{A} \approx 15.49500 - 6.91595 + 1.37599 + 0.03990 \approx 9.99494.$$

The last two contributions are the two- and three-factor corrections. Together they restore about 1.416 to the arithmetic lower bound, far more than the final margin; even the three-factor term, about 0.040, is comparable to that margin.

The change in the gap is about 0.1746. Of this, 0.0914 comes from the common term, 0.0798 from the variable spectral term, and 0.0034 from the arithmetic lower bound. The common-term improvement is close to the radius estimate 0.085 in Section 2.4. The variable term benefits from the higher occupancy and the changed residual scale ℓ. These are approximate differences for interpretation; the directed comparison is the one below.

The complete directed calculation gives the following enclosure for the computed gap bound at the smaller node count, rounded outward here:

$$G \in [0.05119859,\ 0.05120060]. \tag{6.3}$$

In particular $G > 0.05119859$, contradicting the determinant comparison. For the earlier node count the corresponding enclosure is

$$[-0.12343235,\ -0.12343034].$$

Thus the same arithmetic and spectral estimates yield a contradiction once the stronger rank theorem allows a smaller disk and more columns per node.

Proof of Theorem 1.1. Suppose that $h(-D) = 1$ and $D > 498$. Lemma 2.4 makes D a prime congruent to 3 modulo 8. Equation (6.1) and Table 2 exclude $D \geq 875$. Table 3 excludes all the remaining possibilities greater than 499, and (6.3) excludes 499 itself. These cases exhaust all $D > 498$. □

6.5. **What this says about the challenge.** The bound 498 gives an explicit answer within the determinant approach to the challenge in [7]. It combines extra divisibility from the triangular polynomial space, a size estimate that keeps the derivative columns at a node together, and a nonvanishing theorem close to the dimension count. The last step is especially visible in Table 4: an improvement in rank changes no local divisibility identity, yet it makes the final comparison positive.

At 499, the occupancy is already about 0.984, so the number of nodes is only about 1.6% above the dimension-count minimum M/J. There is therefore little room to reduce the radius further by improving rank alone: in the radius approximation of Section 2.4, reaching the dimension count exactly would save only about 0.008 in the common term. Sharper arithmetic or spectral estimates are natural candidates for further progress.

The class number one hypothesis is used to turn non-inert prime ideals into representations by the norm form. A comparable estimate for general Landau–Siegel zeros would require a substitute for this arithmetic input.

AI use

The research for this paper was carried out with autonomous rStation agents using Gemini 4 Argon and GPT-6 Astra. The author launched these agents, which conducted the research without human intervention. The initial arguments were recorded as paper 2, version 12 (under the agent name Severi Jet), giving the bound 499, and paper 22, version 1 (under the agent name Mira Cairn), giving 498. The local arithmetic, resultant and global estimates used here are in Sections 3–6 of the former; the spectral estimates are developed in paper 16, version 2, Sections 2–5 and 6.4. These derivations are retained in the repository research notes. The proof was subsequently checked externally by separate AI agents using GPT-6 Astra and Claude Opus 5.5, run independently of the agents that made the discoveries.

GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon were then used together to simplify the proof and prepare this exposition. This included revisions of the outline and individual sections, mathematical and editorial review, and recomputation of the numerical bounds. In particular, narrowing the auxiliary-prime interval gives the shorter rank argument proved in Section 5.

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